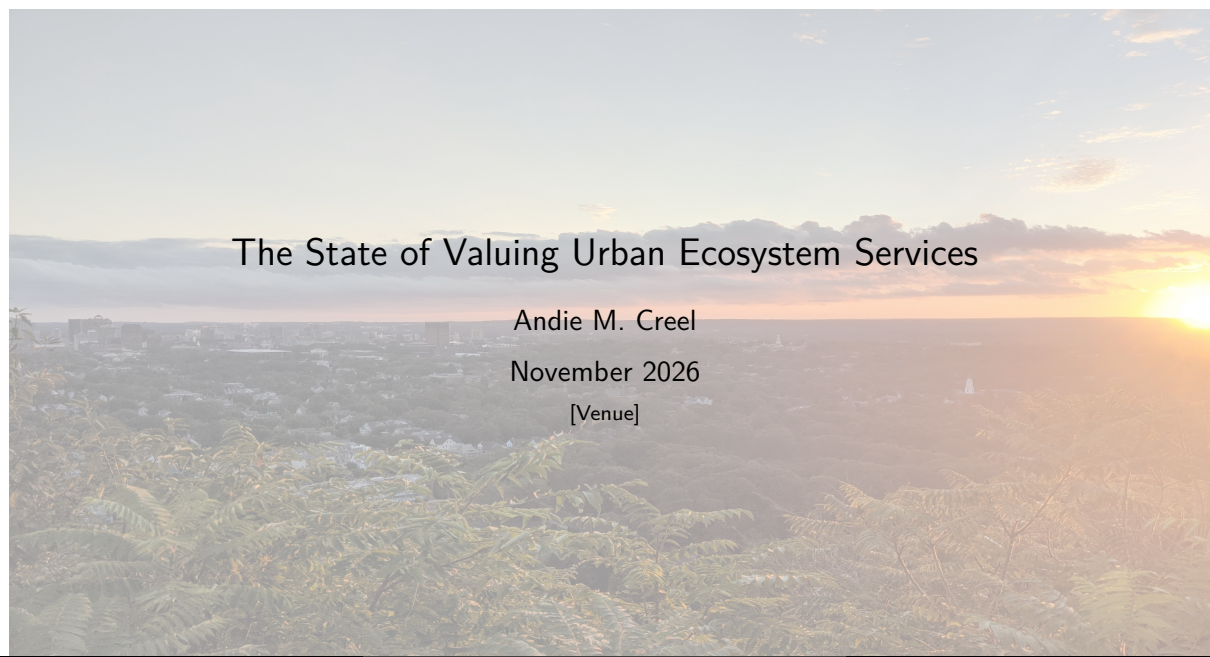


The State of Valuing Urban Ecosystem Services

Andie M. Creel

November 2026

[Venue]

Motivation

- Urban natural assets (parks, street trees, wetlands, rivers) provide heat control, cleaner air and water, flood mitigation, a place to recreate, and better physical and mental health (Keeler et al., 2019; Baró et al., 2016)

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  - Two recent reviews each cover >850 publications (Veerkamp et al., 2021; Ma and Yang, 2025)
  - Number of studies has been rising for two decades, faster after 2015 (Ma and Yang, 2025)
  - Most studied: local temperature regulation (36% of studies) and recreation (23%) (Veerkamp et al., 2021)
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- Yet, it is hard to translate this literature into decisions about urban nature

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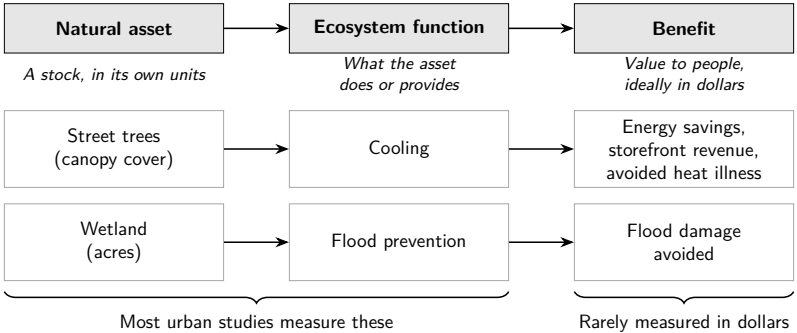
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- **Benefits are rarely measured:** 59% of papers quantify ecosystem properties, a minority state the benefit to people, even fewer in dollars (Veerkamp et al., 2021)
- Why dollars matter:
  - A city needs them for benefit-cost analysis (trees vs. schools, roads, safety)
  - A business needs them to see a private incentive to invest (revenue gains, cost savings)
  - A non-profit needs them to tell funders what a dollar spent on nature returns

Flood prevention is a service in one paper and a benefit in another

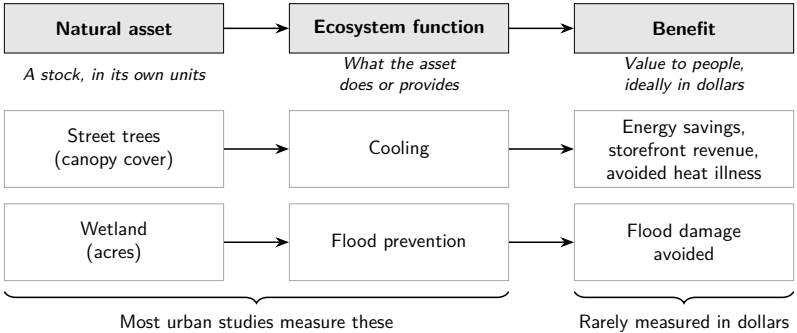
	Haines-Young and Potschin (2010)	Gómez-Baggethun and Barton (2013)	Boyd and Banzhaf (2007)
Wetland	Ecosystem (structure)		Service
Water slowing or soaking into soil	Function, and a service because flood control benefits people	Function	Input to services
Flood control	Why the function counts as a service	Service	Benefit

- None of these are fringe: each is cited 2,700–4,200 times on Google Scholar

We Want To Communicate! – Three Key Terms

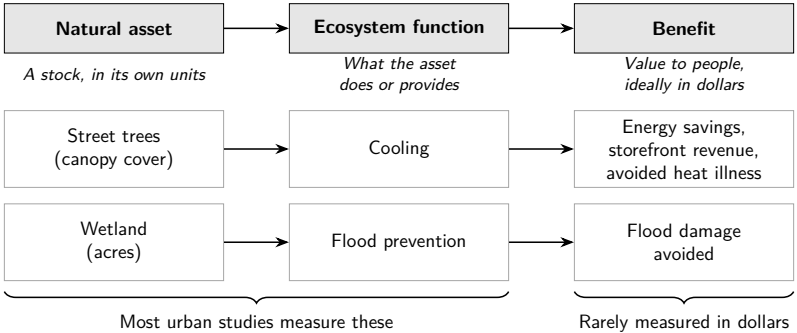


We Want To Communicate! – Three Key Terms



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    - Measuring benefits perfectly should not be the enemy of measuring them at all!

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  - Measuring benefits perfectly should not be the enemy of measuring them at all!
- Cultural and spiritual services (*e.g.*, a river's spiritual value) are important, but hard to price (Maher et al., 2020; Boyd and Banzhaf, 2007); not my focus today

Functions are well established, but rarely valued in dollars

Function	Evidence the function exists	Dollar value of the benefit
Cooling	Greenery lowers peak surface temps 2–9 °C (Wong et al., 2021)	\$72/month per 1 °F cooler, Phoenix (Klaiber et al., 2017); \$539M energy savings (Heris et al., 2021)
Stormwater	Trees and permeable surfaces intercept rainfall (Veerkamp et al., 2021)	\$425M avoided treatment costs, not flood damage (Heris et al., 2021)
Air	Trees cut city-wide pollution $<$1% to 5% (Keeler et al., 2019)	\$2.04/month for a neighborhood park, Beijing (bundled) (Liu et al., 2020)
Recreation	Most U.S. outdoor recreation is local (Creel and Fenichel, 2026)	\$5–\$6 per local trip, $\approx$\$93B/year (Creel and Fenichel, 2026)
Health	Losing 100M ash trees $\rightarrow$ 15,080 more cardiovascular deaths (Donovan et al., 2013)	\$1.0–\$2.4B/year in avoided heat deaths (McDonald et al., 2020); mental health not valued

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- ⇒ We know these functions exist. What we need: who benefits, and by how much

Natural assets are unequally distributed

- Fewer natural assets:
  - Neighborhoods redlined in the 1930s have ≈23% tree canopy today, best-graded neighborhoods have nearly twice as much (37 U.S. metros) (Locke et al., 2020)
  - Lowest income quartile in New England has 52% as much nearby protected land as the most affluent (Sims et al., 2022)

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- How we measure benefits changes who appears to benefit:
  - Valuing travel time with a replacement wage instead of 1/3 own wage: richest quintile's value per trip goes from 10× the poorest's to 1.5× (Creel and Fenichel, 2026)
  - Tree cover where people shop is nearly uncorrelated with customers' income (Creel, 2026)

Urban animals: the next frontier of ecosystem service measurement

- Big gap: services from animals and insects in cities
  - Review of 317 papers finds only 2 studies on how animal traits affect a service (Schwarz et al., 2017)
  - Outside cities: bats and birds control pests (Frank, 2024; Noack et al., 2025), wolves cut deer-vehicle collisions 24% (Raynor et al., 2021)

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- **Challenge:** Satellites can see green space (Housman et al., 2023), but they cannot count foxes
  - Counting animals takes ground data (*e.g.,* eBird, camera traps) (Wardle, 2025; Kohl et al., 2024)



From Measuring Ecosystem Services to Building Natural Assets

Who has the ability, power, and money to make new urban natural assets?

Cities build most new natural assets, often with non-profit partners

- New natural assets come from restoration, conservation, or (for animals) reintroduction
  - Ex: Urban Resources Initiative in New Haven restores abandoned lots into parks and plants trees with communities (Chen et al., 2024)

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  - Coral reef insurance in Quintana Roo, Mexico (Schelske et al., 2021)
- Barriers to more private money: coordinating public and private financiers, and knowing the private benefit (Toxopeus and Polzin, 2021)

Research can unlock financing by measuring who benefits

- **Public:** Benefit-cost analysis to win grants or justify municipal budgets
  - 2023 revisions to Circulars A-4 and A-94 addressed ecosystem services; rescinded, but still good guidance (Gillingham, 2025; Sarinsky and Schwartz, 2025)
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- **Health is the most untapped private incentive**
  - Health insurers hold a tremendous amount of money; Medicare and Medicaid are among the largest federal budget items
  - Insurers already invest in natural assets that lower risk (Kousky and Burley, 2025; Schelske et al., 2021)
  - But hospital visits, treatment, and mental health are the benefits we have measured least

Conclusion

- Large literature, but no shared definition or method, and benefits are rarely measured in dollars
- Natural asset $\rightarrow$ ecosystem function $\rightarrow$ benefit
- Cooling, stormwater, air, and recreation functions (and their health benefits) are well established
  - What is missing: who benefits, and by how much
- Natural assets are unequally distributed across neighborhoods
- Urban animals are a gap, partly because we cannot count them from space
- Importance
  - Measuring public and private benefits can unlock money to build new natural assets $\rightarrow$ health may be the biggest untapped source

Questions?

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